

Collaberry HR Agent — QA Verification Document

Environment: Production
Date of this run: _____
Tester: _____
Client witness: _____

1. Systems under test

Health verified 2026-09-07 09:23 UTC via `railway run --service bot bun scripts/health-check.ts` — all six checks PASS.

Component	URL / Handle	Status
Telegram bot	<code>@Coh3erence_hr_bot</code> (id <code>8811416846</code>)	PASS — process running, polling Telegram
Frontend	<code>https://collaberry-frontend.vercel.app</code>	PASS — 200
Member signup route	<code>https://collaberry-frontend.vercel.app/member-sign-up</code>	PASS — 200
Backend API	<code>https://backend-production-53b74.up.railway.app</code>	PASS — <code>POST /api/users/auth/nonce</code> → 200
Org roster endpoint	<code>GET /api/orgs/{orgId}</code>	PASS — returns org "Aleksa"
Data store	Google Sheet <code>1qM9_Ppm...</code>	PASS — read OK
Language model (Claude)	<code>api.anthropic.com</code>	PASS — 200

Target organisation: "Aleksa", id `63e3ac6c-3e63-4eec-9ac0-de607adf7d05` **Chain:** Arbitrum One (42161) — TeamPoints contract `0x635af529462Fe31cb92C639237207eD7cbAF084e`

Note: the backend root URL returns 404 by design — there is no root route. Health is confirmed via the API endpoints above, not the bare domain.

Participants for this run

Telegram id	Plays	Notes
535329585 (Aleksa)	Candidate + org admin	Runs the section B admin commands before applying; the code excludes a contributor from their own reviewer pool, so he cannot review his own proposal
1971913512 (Simon, @qbfrank)	Reviewer	First half of quorum
302836662 (Gustavo, @sepu85)	Reviewer (client)	Second half of quorum

Reviewer pool is therefore 1971913512 and 302836662 , quorum 2 of 2.

Before you start — both reviewers must have opened the bot. Telegram will not deliver a message to anyone who has never started a chat with the bot, and the failure is silent rather than an error. If either reviewer has not pressed Start at https://t.me/Coh3erence_hr_bot, section E will stall until the 48-hour escalation rather than completing. Confirm both are reachable before beginning.

If one reviewer cannot participate live, run `bun scripts/remove-admin.ts --id <their id>` to reduce the pool to one reviewer (quorum 1) so the session completes deterministically, and restore the row afterwards. Note this makes E4 and E5 not observable — see the note under section E.

2. How to use this document

Each test has a **fixed expected result** agreed in advance. Walk them in order — later tests depend on state created by earlier ones. Record the outcome in the Result column as PASS / FAIL / N-A and put anything surprising in Notes. A FAIL is not necessarily a defect; note it and continue so the session isn't blocked.

3. Configured business rules (verified in code, 2026-08-21)

These are the rules the tests below assert against. All are environment-overridable without a code change.

Rule	Configured value	Source
Reviewer response window	48 hours	REVIEWER_TIMEOUT_HOURS
Max negotiation rounds	2 — enforced	MAX_NEGOTIATION_ROUNDS
Cooldown after rejection	3 days	COOLDOWN_DAYS
Reviewer quorum	Majority — $\text{floor}(n/2)+1$	services/quorum.ts
Hourly → monthly conversion	$\times 160 \text{ h/month}$ ($40\text{h} \times 4\text{wk}$)	FTE_HOURS_PER_MONTH
Fiat portion of comp	0 — remainder in TeamPoints	DEFAULT_FIAT_REQUESTED
Contributor statuses	active , hired , rejected , cooldown	models/types.ts

4. Test cases

A — Access control

ID	Test	Steps	Expected result	Result	Notes
A1	Bot responds	Send <code>/start</code> to <code>@Coh3erence_hr_bot</code>	Bot replies; conversation begins		
A2	Unauthorised user is gated	<code>/start</code> from an account not in <code>AuthorizedUsers</code>	Access is refused, no opportunity data leaks		
A3	Admin authorisation	Admin runs <code>/authorize</code> for a new Telegram id	User is added and can now proceed		

B — Admin opportunity management

Confirms the admin never edits the spreadsheet directly — all changes go through the bot.

ID	Test	Steps	Expected result	Result	Notes
B1	List opportunities	<code>/list_opportunities</code>	Current roles returned with budget and status		
B2	Create opportunity	<code>/add_opportunity</code> , follow prompts	New role appears in B1 and in the sheet		
B3	Edit opportunity	<code>/edit_opportunity</code>	Change persists and is reflected in B1		
B4	Pause opportunity	<code>/pause_opportunity</code>	Role stops being offered to new contributors		

C — Contributor discovery and matching

ID	Test	Steps	Expected result	Result	Notes
C1	Natural-language intake	Describe skills conversationally, not as a form	Bot extracts skills, rate range, availability, timezone without rigid prompts		
C2	Structured capture	Check the Contributors sheet row	Skills, commitment %, rate min/max, timezone, location all populated correctly		
C3	Role match	Continue to matching	Bot proposes a relevant role with a match score		
C4	Poor-fit handling	Offer skills unrelated to any open role	Bot declines gracefully rather than forcing a bad match		

D — Negotiation

ID	Test	Steps	Expected result	Result	Notes
D1	Counter-offer	Propose a rate above budget	Bot negotiates; settlement likelihood reflects distance from budget		
D2	Round cap	Tap "Modify Terms" after a second review	Revision limit reached; only Accept / Walk away are offered — no third round		
D3	Submission	Accept terms	Agreement row created with status submitted; reviewers notified		

E — Review and quorum

ID	Test	Steps	Expected result	Result	Notes
E1	Reviewer notification	Check reviewer account	Reviewer receives the proposal with approve / counter options		
E2	Approve path	Reviewer approves	Decision recorded in ReviewFeedback		
E3	Counter path	Reviewer counters with a different rate	Suggested rate parsed and stored		
E4	Majority quorum	With a pool of n reviewers	Review closes at <code>floor(n/2)+1</code> responses, not unanimity		
E5	Late responder	Reviewer responds after quorum	Late response ignored, outcome unchanged		
E6	48h escalation	Leave a review unanswered past the window	Sweep escalates the agreement (runs every 15 min)		

On E4 and E5. With a two-reviewer pool, `floor(2/2)+1 = 2`, so majority and unanimity coincide and there is no third reviewer left to respond late — neither case is observable in a live run at this pool size. Both are instead evidenced deterministically by the unit tests in `src/services/quorum.test.ts`, which assert the threshold across pool sizes 1–5 (a pool of 3 closing at 2, and of 5 at 3), that a late responder does not change an already-satisfied outcome, and that out-of-pool and repeat votes are not double-counted. Run with `bun test`.

F — Resolution

ID	Test	Steps	Expected result	Result	Notes
F1	Aggregation	Multiple reviewers give differing rates	Feedback aggregated into a single coherent counter-offer		
F2	Reconciliation	Reviewer counter changes the rate	Final agreement terms match the reviewer's counter, not the original ask		
F3	Candidate accepts	Accept the resolved offer	Agreement status → approved		
F4	Candidate rejects	Decline the offer	Contributor enters 3-day cooldown		

G — Collaberry integration (the critical path)

ID	Test	Steps	Expected result	Result	Notes
G1	Invite issued	On acceptance	Bot sends a signup link to <code>/member-sign-up?invitationToken=...</code>		
G2	Signup completes	Open link, connect a fresh wallet, sign up	Account created in Collaberry		
G3	Confirmation tap	Return to Telegram, tap "I've signed up"	Bot links the wallet and creates the on-chain agreement		
G4	Sheet updated	Check Contributors row	status <code>hired</code> , <code>walletAddress</code> and <code>collaberryUserId</code> populated		
G5	Agreement recorded	Check Agreements row	<code>betaAppAgreementId</code> populated		
G6	Renders in app	Open frontend Team view as org admin	Contributor and agreement visible with correct rate and commitment		
G7	Rate conversion	Compare hourly ask to agreement <code>marketRate</code>	$\text{marketRate} = \text{hourly} \times 160$		
G8	Single-use invite	Attempt to reuse a consumed invite token	Second redemption refused cleanly		

5. Known limitations — disclose before testing

These are understood behaviours, not defects discovered during the session. Raising them up front avoids them being logged as surprises.

1. **The "I've signed up" tap is manual.** Nothing polls for signup completion. The contributor must return to Telegram and tap the button after finishing in the browser. Tapping *before* signup completes silently does

nothing — the contributor stays in `active`. This is the most common point of confusion in a live demo.

2. **Each full end-to-end run needs a fresh wallet.** The backend rejects a second agreement for a wallet that already has one. Re-running G with a previously used wallet will fail by design.
3. **TeamPoints balance shows 0 after hiring.** Minting is a separate manual admin action on-chain and is outside the bot's scope.
4. **The bot has no per-message logging.** State is verified by reading the Google Sheet and backend, not by reading logs.
5. **A bot restart loses the conversation in progress.** Conversation state is held in memory, so a redeploy or crash makes the bot forget what it has already collected and ask for it again. Anything already written to the sheet is safe. Do not redeploy during a session.
6. **Data store is a Google Sheet.** Intentional for the MVP so the client can inspect state directly.

6. Pre-go-live items — not blockers for this session

Configuration that is deliberately relaxed for testing and must be tightened before real contributors are onboarded.

Item	State for this session	Required before go-live
<code>NODE_ENV</code>	<code>production</code>	Already done
<code>ALLOW_SELF_REVIEW</code>	<code>false</code> — self-approval is structurally impossible	Already done
Reviewer pool	Two admins (quorum of 2), both reachable	Wider pool as the team grows
Test data	Cleared 2026-09-07 — sheet holds no prior records	Already done

The self-review escape hatch that previously allowed the whole loop to be driven from one Telegram account has been switched off for this session. Every decision below is made by a distinct participant: the candidate cannot approve their own agreement.

7. Outcome

Tests passed: ____ / 32 Failures / follow-ups:

ID	Issue	Severity	Owner	Action

Client sign-off: _____ Date: _____